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基於一致性方法之多詞元預測研究：選擇性分布一致性與潛在一致性匹配

Multi-Token Prediction with Consistency Methods: Selective Distribution Consistency + Latent Consistency Matching

初稿

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摘要

自回歸大型語言模型在推論過程中通常逐一生成詞元，使生成延遲隨輸出序列長度增加。多詞元預測 (Multi-Token Prediction, MTP) 透過共享模型主幹與多個預測頭，使模型能在單次前向傳播中同時預測多個未來詞元，因而具備加速解碼的潛力。然而，標準 MTP 主要由各預測頭針對其對應的真實詞元獨立計算交叉熵損失，缺乏不同預測頭之間的一致性約束，使距離當前位置較遠的未來詞元預測面臨較高的最佳化難度。

為改善上述問題，本研究提出選擇性分布一致性 (Selective Distribution Consistency, SDC)，並結合潛在一致性匹配 (Latent Consistency Matching, LCM)，在不改變 MTP 模型架構的前提下，分別於輸出機率分布與潛在表示層級強化第一預測頭與其餘預測頭之間的一致性。SDC 以第一預測頭作為教師，利用其預測機率分布監督其餘學生預測頭，並透過信心度門檻排除低信心度位置，以降低不可靠教師訊號對一致性學習的影響；LCM 則透過均方誤差縮小教師與學生預測頭之隱藏表示差距。

本研究以 OpenWebText 進行模型訓練，並透過各預測頭之 Oracle perplexity、平均可接受草稿詞元數 (Average Draft Tokens Accepted)、生成文本 perplexity、entropy、4-gram repetition，以及跨資料集零樣本評估，分析所提方法對預測品質、草稿詞元接受能力與生成品質的影響。

正式 1.3B 模型實驗結果顯示，相較於標準 MTP baseline，SDC 與 LCM 聯合訓練可降低四個預測頭的 perplexity，其中第四預測頭的降幅最大，達 5.9%；平均可接受草稿詞元數提升 4.2%；生成文本 perplexity 則降低 52.7%。此外，跨資料集零樣本評估亦顯示，所提方法在多數資料集與預測頭上具有改善效果。整體而言，實驗結果顯示，透過輸出分布與潛在表示兩個層級的一致性學習，可改善較遠未來詞元預測頭的最佳化表現，並提升 MTP 的多詞元預測品質與草稿詞元接受能力，展現進一步提升解碼效率的潛力。

關鍵字：多詞元預測、一致性學習、選擇性分布一致性、潛在一致性匹配、大型語言模型

Abstract

Autoregressive large language models typically generate tokens sequentially during inference, causing generation latency to increase with the length of the output sequence. Multi-Token Prediction (MTP) enables a model to predict multiple future tokens within a single forward pass through a shared model trunk and multiple prediction heads, thereby providing the potential to accelerate decoding. However, standard MTP mainly trains each prediction head independently using the cross-entropy loss associated with its corresponding ground-truth token, without explicitly enforcing consistency across prediction heads. As a result, prediction heads responsible for tokens farther into the future are generally more difficult to optimize.

To address this issue, this study proposes Selective Distribution Consistency (SDC) and combines it with Latent Consistency Matching (LCM). Without modifying the underlying MTP architecture, the proposed approach improves consistency between the first prediction head and the remaining prediction heads at both the output probability distribution and latent representation levels. SDC treats the first prediction head as the teacher and uses its predictive probability distribution to supervise the remaining student heads. A confidence threshold is further applied to exclude low-confidence positions, thereby reducing the influence of unreliable teacher signals during consistency learning. In contrast, LCM minimizes the discrepancy between the hidden representations of the teacher and student heads using the mean squared error.

The models are trained on OpenWebText and evaluated using Oracle perplexity (Oracle PPL) for individual prediction heads, Average Draft Tokens Accepted, generative perplexity (Generative PPL), entropy, 4-gram repetition, and cross-dataset zero-shot evaluation. These metrics are used to assess the effects of the proposed methods on prediction quality, draft-token acceptance, and generation quality.

Experimental results on the main 1.3B-parameter model show that, compared with the standard MTP baseline, joint training with SDC and LCM reduces perplexity across all four prediction heads, with the largest improvement of 5.9% observed for the fourth prediction head. The Average Draft Tokens Accepted increases by 4.2%, while Generative PPL decreases by 52.7%. Cross-dataset zero-shot evaluation further demonstrates improvements across most datasets and prediction heads. Overall, the results indicate that enforcing consistency at both the output distribution and latent representation levels improves the optimization of prediction heads responsible for more distant future tokens, increases draft-token acceptance, and demonstrates the potential for more efficient MTP decoding.

Keywords: Multi-Token Prediction, Consistency Learning, Selective Distribution Consistency, Latent Consistency Matching, Large Language Models

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Chapter 1 Introduction

In recent years, to mitigate the inference latency caused by token-by-token generation in autoregressive large language models, researchers have begun exploring approaches that predict multiple future tokens within a single model inference step. Gloeckle et al.[1] introduced Multi-Token Prediction (MTP), which enables the model to simultaneously predict tokens at multiple future positions during pre-training through a shared model trunk and multiple prediction heads. Subsequently, Samragh et al.[2] further investigated how multi-token prediction capabilities can be introduced into existing pretrained language models through supervised fine-tuning and additional consistency objectives. In addition, DeepSeek-V3[3] incorporated MTP into the pre-training objective of a large-scale language model, demonstrating the practical potential of multi-token prediction for improving inference efficiency.

1.1 Background and Motivation

1.1.1 Background

Large language models typically generate text in an autoregressive manner. Given an input sequence, the model predicts the next token based on the previously observed tokens, appends the newly generated token to the sequence, and then performs the next prediction step. Although this next-token prediction paradigm preserves a clear causal dependency, inference generally requires tokens to be generated sequentially. As a result, the number of forward passes increases with the length of the output sequence.

Multi-Token Prediction (MTP) extends conventional next-token prediction to multiple future positions. An MTP model consists of a shared model trunk and multiple prediction heads. Given the same input representation, the first prediction head predicts the next token, while the remaining prediction heads are responsible for predicting tokens at more distant future positions. Consequently, multiple future-token predictions can be obtained from a single forward pass through the shared model trunk. Since most of the computational cost is concentrated in the shared trunk, while the additional cost introduced by multiple prediction heads is relatively limited, MTP provides the potential to reduce redundant model computation and accelerate decoding.

During inference, the future tokens generated by multiple prediction heads can be treated as draft tokens. If these draft tokens are consistent with the subsequent verification results, multiple tokens can be accepted within a single decoding step, thereby reducing the number of model forward passes required by conventional token-by-token generation. Therefore, the accuracy of future-token prediction and the reliability of the outputs produced by different prediction heads

directly affect the potential decoding efficiency of MTP.

1.1.2 Motivation

Although Multi-Token Prediction (MTP) provides a promising approach for accelerating autoregressive decoding, effectively optimizing multiple prediction heads remains challenging. In standard MTP training, each prediction head is independently optimized using the cross-entropy loss associated with its corresponding ground-truth token. While this objective allows each head to learn future-token prediction, it does not explicitly enforce consistency among the predictions produced by different heads.

As the prediction horizon increases, the uncertainty of future-token prediction also increases. Consequently, prediction heads responsible for more distant future positions are generally more difficult to optimize and tend to exhibit lower prediction accuracy than the first prediction head. Since the effectiveness of multi-token decoding depends on the quality and reliability of the predicted draft tokens, inaccurate predictions from later heads may reduce the number of draft tokens that can be successfully accepted during decoding.

Based on these observations, this study investigates whether additional consistency constraints can improve the optimization of future-token prediction heads without modifying the underlying MTP architecture. Specifically, the first prediction head, which performs next-token prediction, is used as a relatively reliable reference to guide the remaining prediction heads. By encouraging consistency across prediction heads at both the output probability distribution and latent representation levels, this study aims to improve the prediction quality of distant future-token heads and enhance the effectiveness of multi-token decoding.

1.2 Contributions

1. We propose Selective Distribution Consistency for MTP training, which aligns the output distributions of student heads with the first prediction head while filtering low-confidence teacher predictions.
2. We combine SDC with Latent Consistency Matching (LCM) to improve consistency across prediction heads at both the output distribution and latent representation levels without modifying the MTP architecture.
3. Experiments on a 1.3B-parameter model show that SDC+LCM reduces Oracle PPL across prediction heads, increases Average Draft Tokens Accepted, and substantially reduces Generative PPL.

Chapter 2¹ Related Work

2.1 Multi-Token Prediction

⁵² Multi-Token Prediction (MTP) extends conventional next-token prediction by enabling a language model to ¹ predict multiple future tokens simultaneously. Gloeckle et al.[1] proposed an MTP framework consisting of a shared model trunk and multiple prediction heads, where each head is responsible for predicting a different future token. Their results showed that MTP can improve model performance and support self-speculative decoding for faster inference.

Samragh et al.[2] further explored introducing multi-token prediction capabilities into pre-trained autoregressive language models through supervised fine-tuning. Their method incorporates additional training objectives, including consistency learning, to improve the accuracy and coherence of jointly predicted future tokens while enabling faster decoding.

2.2 Speculative Decoding

¹ Stern et al.[4] proposed Blockwise Parallel Decoding, an early approach for accelerating autoregressive generation by predicting multiple future tokens in parallel. The predicted tokens are then validated by a scoring model, and the longest valid prefix is accepted at each decoding step.

Leviathan et al.[5] introduced speculative decoding, in which ³ a lightweight draft model generates multiple candidate tokens that are subsequently verified in parallel by a larger target model. This approach reduces the number of sequential decoding steps while preserving the output distribution of the target model.

Cai et al.[6] proposed Medusa, which ¹ eliminates the need for a separate draft model by adding multiple decoding heads to the target model. These heads ²² predict multiple future tokens in parallel, while a tree-based attention mechanism is used to construct and verify candidate continuations efficiently.

2.3 Consistency Learning

Consistency learning has been widely used as a regularization strategy to improve the stability and generalization of neural networks. ¹⁴ Liang et al.[7] proposed R-Drop, which minimizes the bidirectional KL divergence between the output distributions of two sub-models generated by dropout, thereby encouraging consistent predictions under different stochastic perturbations.

Ni et al.[8] extended consistency regularization from the output level to intermediate representations in Transformer-based language models. Their LR-Drop method aligns hidden states, attention matrices, and output distributions between different dropout-based sub-models, demonstrating that consistency constraints can be applied at multiple representation levels.

For multi-token prediction, Samragh et al.[2] introduced auxiliary consistency objectives to improve the coherence and accuracy of jointly predicted future tokens. Their approach applies consistency learning during supervised fine-tuning, showing that explicitly constraining future-token predictions can improve multi-token generation performance.

2.4 Research Gap

Existing studies have demonstrated the potential of multi-token prediction for improving future-token prediction and accelerating autoregressive decoding. However, standard MTP primarily optimizes each prediction head independently using its corresponding ground-truth token, without explicitly leveraging the more reliable next-token prediction head to guide the optimization of later prediction heads. This issue becomes more significant as the prediction horizon increases, since predicting more distant future tokens is inherently more difficult.

Although previous studies [2] have introduced consistency-based objectives for multi-token prediction, the reliability of the teacher predictions used for consistency supervision has received relatively limited attention. In particular, directly enforcing consistency at low-confidence positions may introduce unreliable supervisory signals. Furthermore, consistency across both output probability distributions and latent representations has not been systematically investigated within the same MTP training framework. These limitations motivate the development of a selective consistency mechanism for improving the optimization of future-token prediction heads.

Chapter 3 Proposed Method

The proposed method introduces consistency objectives into the MTP training process while keeping the model architecture identical to the original MTP design. The overall training objective consists of the original MTP cross-entropy loss, Selective Distribution Consistency (SDC), and Latent Consistency Matching (LCM).

3.1 MTP Training Objective

Given K prediction heads, each head predicts a token at a different future position and is optimized using the cross-entropy loss with its corresponding ground-truth token. The baseline MTP objective is defined as

$$\mathcal{L}_{\text{MTP}} = \sum_{k=1}^K \mathcal{L}_{\text{CE}}^{(k)}, \quad (\text{Equation 3.1})$$

where $\mathcal{L}_{\text{CE}}^{(k)}$ denotes the cross-entropy loss of the k -th prediction head for its corresponding future target token. This objective is retained as the primary token prediction loss in the proposed training framework.

3.2 Selective Distribution Consistency

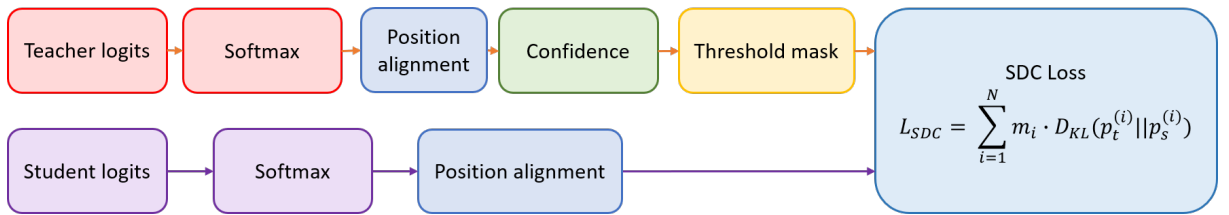


Figure 3.1 : SDC block diagram.

Selective Distribution Consistency (SDC) aligns the output probability distributions of the student heads (Heads 2– K) with the teacher head (Head 1). Since different prediction heads correspond to different future offsets, the teacher and student predictions are first position-aligned such that they correspond to the same target token. Specifically, the prediction of Head k at position i is aligned with the prediction of Head 1 at position $i + k - 1$.

For each token position, the teacher logits are first converted into a probability distribution using softmax:

$$P_t^{(i)} = \text{softmax} \left(z_t^{(i)} \right), \quad (\text{Equation 3.2})$$

where $P_t^{(i)}$ denotes the probability distribution produced by the teacher head (Head 1) at position i .

The confidence of the teacher prediction is then estimated using its normalized entropy:

$$c_i = 1 - \frac{H \left(P_t^{(i)} \right)}{\log(V)}, \quad (\text{Equation 3.3})$$

where $H(\cdot)$ denotes the entropy of the teacher probability distribution and V is the vocabulary size. A higher c_i indicates a more confident teacher prediction.

A confidence threshold τ is subsequently applied to construct a binary mask:

$$m_i = \begin{cases} 1, & \text{if } c_i > \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{Equation 3.4})$$

Thus, only token positions whose teacher confidence exceeds τ participate in consistency learning, while low-confidence positions are excluded from the SDC objective.

For the selected positions, the student probability distribution is encouraged to match the corresponding position-aligned teacher distribution by minimizing the KL divergence:

$$\mathcal{L}_{\text{SDC}}^{(k)} = \frac{1}{N_k} \sum_i m_{i+k-1} D_{\text{KL}} \left(P_t^{(i+k-1)} \parallel P_s^{(i,k)} \right), \quad (\text{Equation 3.5})$$

where $P_t^{(i+k-1)}$ and $P_s^{(i,k)}$ denote the position-aligned teacher and student probability distributions corresponding to the same target token, respectively, and N_k denotes the number of positions selected by the confidence mask for the k -th student head.

The overall SDC objective is defined as

$$\mathcal{L}_{\text{SDC}} = \sum_{k=2}^K \mathcal{L}_{\text{SDC}}^{(k)}. \quad (\text{Equation 3.6})$$

This selective mechanism restricts distribution consistency learning to positions with sufficiently reliable teacher predictions.

Algorithm 3.1: Selective Distribution Consistency

Input: Teacher logits $\{z_t^{(i)}\}$ from Head 1, student logits $\{z_s^{(i,k)}\}_{k=2}^K$, confidence threshold τ

Output: SDC loss $\mathcal{L}_{\text{SDC}}$

$\mathcal{L}_{\text{SDC}} \leftarrow 0$

foreach teacher position i **do**

$P_t^{(i)} \leftarrow \text{softmax}(z_t^{(i)})$

$c_i \leftarrow 1 - \frac{H(P_t^{(i)})}{\log V}$

$m_i \leftarrow \mathbb{I}(c_i > \tau)$

for $k \leftarrow 2$ **to** K **do**

 Align Head k at position i with Head 1 at position $i + k - 1$

$\mathcal{I}_k \leftarrow \{i \mid m_{i+k-1} = 1\}$

$N_k \leftarrow |\mathcal{I}_k|$

$\mathcal{L}_{\text{SDC}}^{(k)} \leftarrow \frac{1}{N_k} \sum_{i \in \mathcal{I}_k} D_{\text{KL}}(P_t^{(i+k-1)} \parallel P_s^{(i,k)})$

$\mathcal{L}_{\text{SDC}} \leftarrow \mathcal{L}_{\text{SDC}} + \mathcal{L}_{\text{SDC}}^{(k)}$

return $\mathcal{L}_{\text{SDC}}$

3.3 Latent Consistency Matching

Latent Consistency Matching (LCM) applies consistency learning at the latent representation level. The same position-alignment strategy used in SDC is applied to the hidden representations before latent consistency matching. Specifically, the hidden representation produced by the k -th student head is aligned with the representation produced by Head 1 that corresponds to the same target token.

For the k -th student head, the LCM loss is defined as

$$\mathcal{L}_{\text{LCM}}^{(k)} = \frac{1}{S_k D} \sum_{i=1}^{S_k} \sum_{d=1}^D \left(h_{t,i,d}^{(k)} - h_{s,i,d}^{(k)} \right)^2, \quad (\text{Equation 3.7})$$

where $h_{t,i,d}^{(k)}$ and $h_{s,i,d}^{(k)}$ denote the position-aligned teacher and student hidden representations, respectively, S_k denotes the number of valid aligned token positions for the k -th student head, and D denotes the hidden dimension.

The overall LCM objective is defined as

$$\mathcal{L}_{\text{LCM}} = \sum_{k=2}^K \mathcal{L}_{\text{LCM}}^{(k)}. \quad (\text{Equation 3.8})$$

While SDC constrains the prediction heads at the output distribution level, LCM provides

an additional consistency constraint on their position-aligned latent representations.

3.4 Joint Training Objective ⁵³

The final training objective combines the original MTP loss with the two consistency objectives:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MTP}} + \lambda_{\text{SDC}}\mathcal{L}_{\text{SDC}} + \lambda_{\text{LCM}}\mathcal{L}_{\text{LCM}}, \quad (\text{Equation 3.9})$$

where λ_{SDC} and λ_{LCM} control the contributions of SDC and LCM, respectively.

During training, the teacher outputs are detached when computing the consistency objectives. Therefore, SDC and LCM use Head 1 as the reference to optimize Heads 2– K , while the original cross-entropy objective continues to optimize all prediction heads.



Chapter 4 Experimental Results

4.1 Experimental Setup and Evaluation Metrics

The main experiments were conducted using a 1.3B-parameter MTP model with four prediction heads. The models were trained on OpenWebText[9]. The baseline and proposed models were trained on 10 billion tokens, while the ablation experiments used models trained on 1 billion tokens. The main SDC+LCM configuration used $\lambda_{\text{SDC}} = 1.0$, $\tau = 0.3$, and $\lambda_{\text{LCM}} = 0.5$. Detailed training configurations are provided in Appendix A.

The baseline model was optimized using the original MTP training objective, whereas the proposed model additionally incorporated Selective Distribution Consistency (SDC) and Latent Consistency Matching (LCM). Both settings used the same MTP architecture, allowing the effects of the consistency objectives to be evaluated independently of architectural changes.

Unless otherwise specified, values reported with $\pm$ denote the mean $\pm$ standard deviation over three random seeds.

The models were evaluated in terms of prediction quality, draft-token acceptance, generation quality, and cross-dataset generalization. The corresponding evaluation procedures and results are described in the following sections.

4.2 Main Experimental Results

4.2.1 Oracle Perplexity

Oracle PPL was computed independently for each prediction head using its corresponding ground-truth future token, where lower PPL indicates better prediction quality.

Table 4.1 : Oracle PPL comparison between the baseline and SDC+LCM.

Setting	Head 1	Head 2	Head 3	Head 4
Baseline	19.4293	87.2351	200.0813	318.3247
SDC+LCM	19.1695	83.7266	189.2825	299.5625
Improvement	1.34%	4.02%	5.40%	5.89%

Table 4.1 compares the Oracle PPL of the baseline MTP model and the proposed SDC+LCM model across four prediction heads. The proposed method consistently reduces PPL for all prediction heads. Head 1 shows a relatively small improvement of 1.34%, whereas larger improve-

ments are observed for the later prediction heads. Specifically, the PPL reductions for Heads 2, 3, and 4 are 4.02%, 5.40%, and 5.89%, respectively.

The improvement becomes more pronounced as the prediction position moves farther into the future, with Head 4 achieving the largest reduction. This result indicates that consistency-based training is particularly beneficial for the optimization of later prediction heads, which are responsible for more distant future-token predictions. Although the consistency objectives primarily constrain Heads 2–4, a slight improvement is also observed for Head 1.

4.2.2 Draft Token Acceptance

Table 4.2 : Comparison of Average Draft Tokens Accepted between the baseline and SDC+LCM.

Setting	Overall	Prefix 128	Prefix 256	Prefix 512	Prefix 768
Baseline	0.7010	0.9117 ± 0.0205	0.9458 ± 0.0068	0.9689 ± 0.0221	0.9760 ± 0.0222
SDC+LCM	0.7302	0.9653 ± 0.0401	0.9873 ± 0.0063	1.0158 ± 0.0309	1.0272 ± 0.0250
Improvement	4.17%	5.88%	4.39%	4.84%	5.25%

Draft-token acceptance was evaluated using greedy predictions from the MTP heads. Head 1 was used as the verification reference, while predictions from Heads 2– K were treated as draft tokens. Draft tokens were accepted sequentially when they matched the corresponding Head 1 predictions, and verification stopped at the first mismatch. Head 1 itself was not included in the accepted draft-token count.

Average Draft Tokens Accepted denotes the average number of consecutively accepted draft tokens per evaluation instance. The overall result was computed over all valid positions, while additional evaluations were conducted using random contiguous prefixes of lengths 128, 256, 512, and 768.

Table 4.2 compares the Average Draft Tokens Accepted between the baseline and SDC+LCM. The proposed method increases the overall Average Draft Tokens Accepted from 0.7010 to 0.7302, corresponding to an improvement of 4.17%.

Consistent improvements are also observed across different prefix lengths. At prefix lengths of 128, 256, 512, and 768, SDC+LCM achieves improvements of 5.88%, 4.39%, 4.84%, and 5.25%, respectively. Among these settings, the largest improvement is observed at a prefix length of 128, while the highest absolute number of accepted draft tokens is achieved at a prefix length of 768.

These results indicate that the proposed consistency objectives improve the reliability of future-token predictions, resulting in more draft tokens being accepted under the self-speculative decoding criterion.

4.2.3 Generative Quality

Generation quality was evaluated using Generative PPL, entropy, and 4-gram repetition. A total of 200 samples were generated using the first 128 tokens of each evaluation sequence as the prompt. Generation was performed using top- p and temperature sampling with $p = 0.9$ and a temperature of 0.8. Generative PPL was computed using GPT-2 Large as an external language model, where only the generated continuation was included in the perplexity calculation. Entropy was used to characterize the diversity of generated tokens, while 4-gram repetition was used to measure repetitive generation patterns.

Table 4.3 : Comparison of generative quality between the baseline and SDC+LCM.

Setting	Generative PPL	Entropy	4-gram Repetition
Baseline	39.5580 ± 0.9425	7.3223 ± 0.0351	0.0229 ± 0.0084
SDC+LCM	18.7180 ± 0.8276	6.8620 ± 0.0202	0.0526 ± 0.0147
Change	-52.68%	-6.29%	+130.17%

Table 4.3 compares the generation quality of the baseline and SDC+LCM models. The proposed method substantially reduces Generative PPL from 39.5580 to 18.7180, corresponding to a reduction of 52.68%. This result indicates that the generated text is evaluated as more predictable by the external language model.

In addition, the entropy decreases from 7.3223 to 6.8620, representing a reduction of 6.29%. However, the 4-gram repetition increases from 0.0229 to 0.0526, corresponding to a relative increase of 130.17%.

These results reveal a trade-off in generation behavior. Although SDC+LCM significantly improves Generative PPL, the lower entropy and higher 4-gram repetition indicate reduced generation diversity and an increased tendency toward repetitive patterns. Nevertheless, the absolute increase in 4-gram repetition is approximately 0.03.

4.3 Zero-Shot Evaluation

Table 4.4 : Zero-shot Oracle PPL improvement of SDC+LCM over the baseline across different datasets.

Dataset	Head 1	Head 2	Head 3	Head 4
WikiText	0.32%	7.57%	7.19%	5.62%
PTB	13.82%	9.90%	10.53%	5.63%
LAMBADA	38.53%	10.43%	4.44%	3.27%
arXiv	1.48%	3.94%	3.71%	0.27%
PubMed	7.04%	7.76%	8.33%	8.26%

To evaluate the generalization ability of the proposed method across different domains, zero-shot evaluation was conducted on WikiText[10], Penn Treebank (PTB)[11], LAMBADA[12],

and the arXiv and PubMed subsets of the Scientific Papers dataset[13]. Table 4.4 reports the relative Oracle PPL improvement of SDC+LCM over the baseline for each prediction head.

Overall, SDC+LCM achieves PPL improvements across all evaluated datasets and prediction heads, although the magnitude of improvement varies considerably across domains. For Head 1, relatively large improvements are observed on PTB[11], LAMBADA[12], and PubMed[13], with LAMBADA[12] showing the largest reduction of 38.53%. In contrast, the improvement on WikiText[10] is limited to 0.32%.

For WikiText[10], the improvements are mainly concentrated on Heads 2–4, with reductions of 7.57%, 7.19%, and 5.62%, respectively, while the improvement for Head 1 is relatively limited.

The improvement patterns also vary for later prediction heads across different domains. For example, Head 4 achieves an 8.26% improvement on PubMed but only a 0.27% improvement on arXiv. These results indicate that the effectiveness of consistency-based training depends partly on the characteristics of the evaluation domain, while generally yielding improvements across unseen datasets.

4.4 Ablation Study

4.4.1 Overall Ablation Results

Table 4.5 : Ablation results of Oracle PPL under different consistency settings.

Setting	Head 1	Head 2	Head 3	Head 4
Baseline	26.2356	121.8033	269.2450	406.6959
$\lambda_{SDC} = 0.5, \tau = 0.3, \lambda_{LCM} = 0$	26.4587	120.7186	265.0660	402.0413
$\lambda_{SDC} = 1.5, \tau = 0.3, \lambda_{LCM} = 0$	27.2684	122.5244	264.6119	397.3106
$\lambda_{SDC} = 1.0, \tau = 0.0, \lambda_{LCM} = 0$	26.7541	120.6687	261.6664	394.6503
$\lambda_{SDC} = 1.0, \tau = 0.3, \lambda_{LCM} = 0$	26.8150	120.5276	261.9479	394.7055
$\lambda_{SDC} = 1.0, \tau = 0.6, \lambda_{LCM} = 0$	27.7396	129.2922	278.8881	417.5874
$\lambda_{SDC} = 0, \tau = 0, \lambda_{LCM} = 0.5$	25.7096	118.9247	262.1637	397.9911
$\lambda_{SDC} = 0, \tau = 0, \lambda_{LCM} = 1.0$	25.5312	118.5823	261.1872	395.1838
$\lambda_{SDC} = 0, \tau = 0, \lambda_{LCM} = 1.5$	25.4583	119.1595	263.2083	398.8679
$\lambda_{SDC} = 1.0, \tau = 0.3, \lambda_{LCM} = 0.5$	26.2402	118.2168	257.0344	387.7455
$\lambda_{SDC} = 1.0, \tau = 0.3, \lambda_{LCM} = 1.0$	26.4553	119.2975	258.4100	390.7829
$\lambda_{SDC} = 1.5, \tau = 0.3, \lambda_{LCM} = 1.0$	26.5485	118.4341	254.9612	383.7626

Bold values indicate the best result in each column.

Table 4.6 : Ablation results of Average Draft Tokens Accepted under different consistency settings.

Setting	Overall	Prefix 128	Prefix 256	Prefix 512	Prefix 768
Baseline	0.6238	0.8251 $\pm$ 0.0140	0.8205 $\pm$ 0.0062	0.8519 $\pm$ 0.0195	0.8552 $\pm$ 0.0233
$\lambda_{SDC} = 0.5, \tau = 0.3, \lambda_{LCM} = 0$	0.6345	0.8195 $\pm$ 0.0122	0.8416 $\pm$ 0.0115	0.8601 $\pm$ 0.0065	0.8644 $\pm$ 0.0304
$\lambda_{SDC} = 1.5, \tau = 0.3, \lambda_{LCM} = 0$	0.6414	0.8522 $\pm$ 0.0098	0.8581 $\pm$ 0.0139	0.8897 $\pm$ 0.0043	0.9082 $\pm$ 0.0204
$\lambda_{SDC} = 1.0, \tau = 0.0, \lambda_{LCM} = 0$	0.6388	0.8374 $\pm$ 0.0165	0.8746 $\pm$ 0.0108	0.8736 $\pm$ 0.0158	0.8802 $\pm$ 0.0029
$\lambda_{SDC} = 1.0, \tau = 0.3, \lambda_{LCM} = 0$	0.6400	0.8406 $\pm$ 0.0131	0.8602 $\pm$ 0.0232	0.8720 $\pm$ 0.0233	0.8772 $\pm$ 0.0015
$\lambda_{SDC} = 1.0, \tau = 0.6, \lambda_{LCM} = 0$	0.6113	0.7908 $\pm$ 0.0074	0.7919 $\pm$ 0.0086	0.8009 $\pm$ 0.0035	0.8174 $\pm$ 0.0130
$\lambda_{SDC} = 0, \tau = 0, \lambda_{LCM} = 0.5$	0.6398	0.8517 $\pm$ 0.0156	0.8589 $\pm$ 0.0096	0.8794 $\pm$ 0.0050	0.8895 $\pm$ 0.0257
$\lambda_{SDC} = 0, \tau = 0, \lambda_{LCM} = 1.0$	0.6442	0.8491 $\pm$ 0.0130	0.8698 $\pm$ 0.0154	0.8919 $\pm$ 0.0168	0.8943 $\pm$ 0.0012
$\lambda_{SDC} = 0, \tau = 0, \lambda_{LCM} = 1.5$	0.6467	0.8510 $\pm$ 0.0112	0.8862 $\pm$ 0.0054	0.8951 $\pm$ 0.0129	0.9028 $\pm$ 0.0327
$\lambda_{SDC} = 1.0, \tau = 0.3, \lambda_{LCM} = 0.5$	0.6507	0.8498 $\pm$ 0.0339	0.8615 $\pm$ 0.0179	0.8987 $\pm$ 0.0103	0.8862 $\pm$ 0.0151
$\lambda_{SDC} = 1.0, \tau = 0.3, \lambda_{LCM} = 1.0$	0.6495	0.8402 $\pm$ 0.0209	0.8612 $\pm$ 0.0043	0.8879 $\pm$ 0.0157	0.8950 $\pm$ 0.0071
$\lambda_{SDC} = 1.5, \tau = 0.3, \lambda_{LCM} = 1.0$	0.6525	0.8590 $\pm$ 0.0101	0.8656 $\pm$ 0.0214	0.9023 $\pm$ 0.0277	0.9044 $\pm$ 0.0140

⁴ Bold values indicate the best result in each column.

Table 4.7 : Ablation results of generative quality under different consistency settings.

Setting	Generative PPL	Entropy	4-gram Repetition
Baseline	83.8285 $\pm$ 5.1409	7.4068 $\pm$ 0.0461	0.0119 $\pm$ 0.0097
$\lambda_{SDC} = 0.5, \tau = 0.3, \lambda_{LCM} = 0$	52.7654 $\pm$ 2.2732	7.1490 $\pm$ 0.0335	0.0323 $\pm$ 0.0169
$\lambda_{SDC} = 1.5, \tau = 0.3, \lambda_{LCM} = 0$	31.8082 $\pm$ 0.4503	6.8248 $\pm$ 0.0353	0.0485 $\pm$ 0.0159
$\lambda_{SDC} = 1.0, \tau = 0.0, \lambda_{LCM} = 0$	68.0596 $\pm$ 4.5760	7.3141 $\pm$ 0.0519	0.0193 $\pm$ 0.0154
$\lambda_{SDC} = 1.0, \tau = 0.3, \lambda_{LCM} = 0$	48.4875 $\pm$ 1.2280	7.1184 $\pm$ 0.0336	0.0183 $\pm$ 0.0055
$\lambda_{SDC} = 1.0, \tau = 0.6, \lambda_{LCM} = 0$	28.2332 $\pm$ 1.2754	6.7789 $\pm$ 0.0178	0.0393 $\pm$ 0.0080
$\lambda_{SDC} = 0, \tau = 0, \lambda_{LCM} = 0.5$	26.2091 $\pm$ 1.5675	6.8576 $\pm$ 0.0374	0.0555 $\pm$ 0.0204
$\lambda_{SDC} = 0, \tau = 0, \lambda_{LCM} = 1.0$	24.0955 $\pm$ 3.2440	6.7935 $\pm$ 0.0828	0.0657 $\pm$ 0.0388
$\lambda_{SDC} = 0, \tau = 0, \lambda_{LCM} = 1.5$	29.9033 $\pm$ 0.5373	6.8644 $\pm$ 0.0249	0.0215 $\pm$ 0.0049
$\lambda_{SDC} = 1.0, \tau = 0.3, \lambda_{LCM} = 0.5$	21.4788 $\pm$ 0.0773	6.6876 $\pm$ 0.0421	0.0720 $\pm$ 0.0062
$\lambda_{SDC} = 1.0, \tau = 0.3, \lambda_{LCM} = 1.0$	26.3414 $\pm$ 0.3656	6.7567 $\pm$ 0.0016	0.0460 $\pm$ 0.0160
$\lambda_{SDC} = 1.5, \tau = 0.3, \lambda_{LCM} = 1.0$	25.7832 $\pm$ 1.6338	6.7177 $\pm$ 0.0455	0.0453 $\pm$ 0.0197

Bold indicates the lowest Generative PPL.

4.4.2 Effect of SDC Weight

To investigate the effect of the SDC loss weight, the confidence threshold and LCM weight were fixed at $\tau = 0.3$ and $\lambda_{LCM} = 0$, respectively, while the SDC weight was varied among $\lambda_{SDC} = 0.5, 1.0$, and 1.5 . The baseline without consistency regularization is also included for comparison.

Increasing the SDC weight generally improves draft-token acceptance and reduces Generative PPL, while the Oracle PPL results indicate that excessively strong distribution consistency does not consistently benefit all prediction heads.

4.4.3 Effect of Confidence Threshold

To investigate the effect of the confidence threshold, $\lambda_{SDC} = 1.0$ and $\lambda_{LCM} = 0$ were fixed while τ was varied among $0.0, 0.3$, and 0.6 .

The Oracle PPL results are similar between $\tau = 0.0$ and $\tau = 0.3$, whereas $\tau = 0.6$ significantly degrades all prediction heads. A similar trend is observed for Average Draft Tokens Accepted, which decreases from 0.6400 at $\tau = 0.3$ to 0.6113 at $\tau = 0.6$.

In contrast, Generative PPL continues to decrease as τ increases, but this is accompanied by lower entropy and higher 4-gram repetition. Therefore, an excessively high confidence threshold may improve Generative PPL while degrading future-token prediction and draft-token acceptance, indicating a trade-off among evaluation metrics.

4.4.4 Effect of LCM Weight

To investigate the effect of the LCM weight, $\lambda_{SDC} = 0$ was fixed while λ_{LCM} was varied among $0.5, 1.0$, and 1.5 .

Increasing λ_{LCM} from 0.5 to 1.0 generally improves Oracle PPL across the prediction heads. However, further increasing λ_{LCM} to 1.5 slightly degrades the PPL of Heads 2–4, suggesting that excessive latent consistency does not provide additional benefits.

Average Draft Tokens Accepted gradually increases from 0.6398 to 0.6467 as λ_{LCM} increases. In contrast, Generative PPL reaches its lowest value at $\lambda_{LCM} = 1.0$ and increases again at $\lambda_{LCM} = 1.5$. These results indicate that a moderate LCM weight provides a better balance between prediction quality and draft-token acceptance.

4.4.5 Joint Effect of SDC and LCM

To evaluate the joint effect of SDC and LCM, configurations using both consistency objectives were compared with SDC-only and LCM-only settings.

Joint training generally achieves lower Oracle PPL for Heads 2–4 and higher Average Draft

Tokens Accepted than using either consistency objective alone. In particular, the joint configurations achieve draft acceptance values of approximately 0.65, compared with 0.6400 for the SDC-only setting and 0.6442 for the LCM-only setting.

For generation quality, joint training further reduces Generative PPL, but is accompanied by lower entropy and higher 4-gram repetition. Therefore, SDC and LCM can complement each other in improving future-token prediction and draft-token acceptance, while a trade-off with generation diversity remains.

4.5 Discussion

⁴⁷ The experimental results show that consistency-based training is particularly beneficial for prediction heads responsible for more distant future tokens. Across the main experiments, SDC+LCM consistently reduces Oracle PPL, with larger improvements generally observed for the later prediction heads.

The improvement in future-token prediction is also reflected in draft-token acceptance. SDC+LCM increases the Average Draft Tokens Accepted, while the zero-shot results indicate that the benefits of consistency-based training can generalize across different domains.

However, the ablation study shows that stronger consistency constraints do not always lead to better overall performance. In particular, lower Generative PPL is often accompanied by lower entropy and higher repetition, indicating a trade-off between prediction consistency and generation diversity. Therefore, the strengths of SDC, LCM, and the confidence threshold should be carefully balanced.

Chapter 5 Conclusion and Future Work

5.1 Conclusion

This thesis investigates consistency-based training for Multi-Token Prediction (MTP) to improve the optimization of future-token prediction heads. Two consistency objectives are introduced: Selective Distribution Consistency (SDC), which selectively aligns the output distributions of student heads with the first prediction head, and Latent Consistency Matching (LCM), which aligns their latent representations. Both objectives are incorporated into training without modifying the original MTP architecture.

Experimental results on the 1.3B-parameter model demonstrate that SDC+LCM improves future-token prediction and draft-token acceptance. Compared with the baseline, Oracle PPL is reduced across all four prediction heads, with Head 4 achieving the largest improvement of 5.89%. The Average Draft Tokens Accepted increases from 0.7010 to 0.7302, corresponding to an improvement of 4.17%. In addition, Generative PPL decreases from 39.5580 to 18.7180, representing a reduction of 52.68%. Zero-shot evaluation further shows improvements across multiple unseen datasets.

The ablation study shows that the strengths of SDC and LCM, as well as the confidence threshold, should be carefully balanced. Lower Generative PPL is often accompanied by lower entropy and higher repetition, indicating a trade-off with generation diversity. Overall, the proposed consistency-based training framework improves the prediction quality of MTP heads, particularly for more distant future-token predictions, while also increasing draft-token acceptance without modifying the model architecture.

5.2 Future Work

Several directions can be explored in future work. First, the proposed consistency objectives can be evaluated with larger models and larger training budgets. Since the ablation study in this work was conducted using models trained on only 1 billion tokens, larger-scale experiments may provide a more comprehensive understanding of the effects of SDC, LCM, and their associated hyperparameters.

Second, future studies can investigate the application of SDC and LCM during supervised fine-tuning (SFT). This would help determine whether the proposed consistency objectives remain effective when adapted to downstream instruction-following and generation tasks.

Third, future work should evaluate the actual inference speed-up of the proposed method using a complete self-speculative decoding system. Although this work demonstrates an in-

crease in Average Draft Tokens Accepted, practical acceleration also depends on factors such as verification overhead, memory access, and hardware utilization. Therefore, wall-clock latency and end-to-end decoding throughput should be measured to determine the practical efficiency gains of the proposed approach.

Finally, the trade-off between consistency and generation diversity deserves further investigation. Future work may explore alternative regularization strategies or adaptive consistency weighting to preserve the improvements in prediction quality while mitigating the increase in repetitive generation.



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Appendix A Experimental Configuration

For optimization, AdamW [14] was used with $\beta_1 = 0.9$, $\beta_2 = 0.95$, and a decoupled weight decay coefficient of 0.1. The learning rate was linearly warmed up, followed by a cosine decay schedule [15]. In addition, gradients were clipped to a maximum Euclidean norm of 1.0 throughout training.

Table A.1 : Detailed experimental configuration.

Configuration	Value
Model Size	1.3B
Number of Prediction Heads	4
Main Training Tokens	10B
Ablation Training Tokens	1B
Sequence Length	1024
Batch Size	8
Optimizer	AdamW
Peak LR	3×10^{-4}
Weight Decay	0.1
Warm-up Steps	15000
λ_{SDC}	1.0
Confidence Threshold τ	0.3
λ_{LCM}	0.5
Top- p	0.9
Temperature	0.8

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